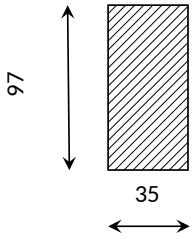
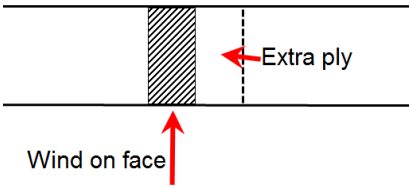
Plated spandrel panel stud design check to EC5 (Flatwise timber):





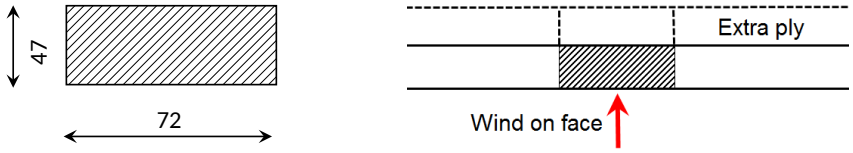
Roof pitch:		30 deg			
Roof load:	dead	0.685 kN/m²	Site snow:	0.480 kN/m²	
Gable ladder overhang:		0.600 m	Distance to 1st truss	0.050 m	
Dynamic wind pressure:		0.60 kN/m²			
Wall type					
		self wt	0.100 kN/m²		
max stud length		3.400 m	stud c/c	0.60 m	
Timber	35 mm (Width)	97 mm (Depth)		1 ply	

Check gable panel for wind on face as leading action combined with vertical loads
PL NA.1: short-term wind action; confirm external and internal wind coefficients.
NZ = unverified assumptions; scope and missing PN-EN 1990/NA: see documentation.

Design axial stress	0.254 N/mm²	Design bending stress	16.349 N/mm²
Design axial strength	0.432 N/mm²	Design bending strength	18.129 N/mm²
kc,min	0.030		
csi	1.490	FAIL	
deflection	24.91 mm	allowable deflection	15.3 mm FAIL

Deflection limit is a project assumption, not a verified PL limit for studs.

Plated spandrel panel stud design check to EC5 (Edgewise timber):



Roof pitch:		30	deg		
Roof load:	dead	0.685	kN/m²	Site snow:	0.480 kN/m²
Gable ladder overhang:		0.600	m	Distance to 1st truss	0.050 m
Dynamic wind pressure:		0.600	kN/m²		
Wall type:					
		self wt		0.240	kN/m²
max stud length		3.60	m	stud c/c	0.60 m
Multi-ply fixing	nail dia	3.35	mm	nail spacing	300 mm
Timber		72	mm	47	mm ply
		Width		Depth	

Check gable panel for wind on as leading action face combined with vertical loads
PL NA.1: short-term wind action; confirm external and internal wind coefficients.
NZ = unverified combination / model / fasteners; see documentation.

Design axial stress	0.130	N/mm²	Design bending stress	9.720	N/mm²
Design axial strength	0.688	N/mm²	Design bending strength	20.956	N/mm²
kc,min	0.047				
csi	2.026	FAIL			
deflection	26.77	mm	allowable deflection	16.2 mm	FAIL

Deflection limit is a project assumption, not a verified PL limit for studs.
Calculation of effective EI and resulting bending stress for two or three members nailed together
PN-EN 1995-1-1:2010 Annex B; fastener resistance must be verified separately.